CL3001

Machine Learning - Lab

Semester Project (Spring 2026)

Deadline:29-November 2026

#CLO5: Develop a project using machine learning techniques learned during the course.

1. Introduction

You will choose a recent research paper (published in **2025 or 2026**), understand the problem it solves, rebuild its experiment using the **same dataset and the same methodology**, and then **compare your results with the results reported in the paper**.

2. Learning Objectives

After completing this project you will be able to:

1. Read and summarize a scientific paper critically.
2. Identify a problem statement, objectives and contributions.
3. Prepare and explore a dataset.
4. Implement a published methodology using techniques taught in this course.
5. Evaluate results with proper metrics and compare them with published results.
6. Explain differences, limitations and possible improvements.
7. Present your work in a clear written report and a short presentation.

3. Project Overview (What You Must Do)

The project has seven steps:

1. Choose a paper (2025 or 2026).
2. Identify the problem statement.
3. Get the dataset used by the paper.
4. Apply the paper's methodology.
5. Record your results.
6. Compare your results with the paper's results.

7. Write the report and present.

4. Step-by-Step Instructions

Step 1: Choose a Paper

You are free to choose **any paper of your own choice**, as long as it meets all the rules below.

Selection rules

- The paper must be published in **2025 or 2026** (journal, conference or arXiv with a clear venue/date).
- The topic must be one **taught in this course**. The methods in the paper must be methods you have studied in class.
- The paper must be **application-based**, meaning it solves a real-world problem (for example, healthcare, agriculture, finance, education, traffic, security, e-commerce, environment).
- The paper must use a **publicly available dataset** (Kaggle, UCI, Hugging Face, GitHub, government or university repositories, etc.).
- The paper must clearly report **numerical results** (accuracy, F1-score, RMSE, etc.) so that you can compare.
- The methodology must be described clearly enough to be re-implemented with your available computer resources (free Google Colab or Kaggle notebooks are acceptable).

Where to find papers: Google Scholar, IEEE Xplore, ScienceDirect, SpringerLink, MDPI, ACM Digital Library.

Tip: Search with keywords such as *"your topic + application + 2025"*. Choose a paper that is neither too simple (a single basic model) nor too heavy (needs huge GPUs or weeks of training).

Approval: Submit the paper title, link, dataset link and a short note on why you chose it. Two students in the same group should not choose the same paper unless the instructor allows it.

Step 2: Identify the Problem Statement

Read the paper carefully (abstract, introduction, methodology, results, conclusion). Then write, in your own words:

- **Application area:** where is this problem found in real life?
- **Problem statement:** what exactly is the paper trying to solve, in 3 to 5 sentences?
- **Motivation:** why is it important?
- **Existing limitations:** what did earlier work fail to do?
- **Proposed solution:** what idea does the paper offer?

- **Objectives and contributions** of the paper.

Example: *"Early detection of plant leaf diseases is difficult for farmers without experts. The paper proposes a CNN-based classifier trained on a leaf image dataset to identify disease categories automatically with high accuracy."*

Step 3: Understand and Prepare the Dataset

Use **the same dataset** that the paper used. Document:

- Dataset name, source link and size (number of samples, features, classes).
- Description of features/columns and the target variable.
- Class distribution (balanced or imbalanced).
- Data quality issues (missing values, duplicates, outliers, noise).
- Pre-processing done in the paper (cleaning, normalization, encoding, resizing, tokenization, feature selection, train/validation/test split, etc.).

Follow the **same pre-processing and the same split ratio** as the paper. If the paper does not mention something, make a sensible choice and write it as an **assumption**.

Include exploratory data analysis (EDA): tables, charts and short comments.

Step 4: Apply the Paper's Methodology

Implement the **same pipeline** as the paper:

- Same model(s) or algorithm(s).
- Same hyperparameters where stated (learning rate, number of layers, epochs, k value, kernel, etc.).
- Same evaluation protocol (hold-out split or k-fold cross-validation).
- Same metrics.

Rules for implementation:

- Write your **own code** (Python recommended). Using standard libraries (NumPy, pandas, scikit-learn, TensorFlow, PyTorch) is allowed.
- Official code from the paper's authors may be **read for understanding only**; you may not copy it as your submission. If you use any external code, you must cite it and explain what you changed.
- Fix a random seed so that your results are reproducible.
- Keep a log of everything you change.

Step 5: Record Your Results

Report your results using the same metrics as the paper, for example Accuracy, Precision, Recall, F1-score, ROC-AUC, confusion matrix, MAE, RMSE, R^2. Add training curves or plots where relevant. Run the experiment more than once (for example 3 to 5 seeds) and report the mean, and standard deviation if possible.

Step 6: Compare with the Paper's Results

Create a comparison table like this:

Metric	Paper's Result	Your Result	Difference
Accuracy			
F1-score			
Other			

Then answer these questions in writing:

1. Did you match the paper's results? If yes, how closely?
2. If your results are different (higher or lower), what are the **possible reasons**? (for example different data split, missing hyperparameters, random seed, library versions, pre-processing choices, hardware limits)
3. Was anything unclear or missing in the paper that made replication hard?
4. Do you find any weakness in the paper (small dataset, data leakage, unfair comparison, missing details)?

Important: You will **not lose marks** because your results are lower than the paper's. You lose marks only for hiding differences or reporting fake numbers. Honest analysis is what earns marks.

Step 7: Optional Improvement (Bonus)

For bonus marks you may try one small improvement (a different hyperparameter setting, another model from the course, class balancing, extra feature engineering) and compare it against both the paper and your replication.

5. Topics Allowed

Your paper's methodology must belong to the topics covered in this course.
Example of how a list may look for a machine learning course: data pre-processing, regression, classification (KNN, Decision Tree, SVM, Naive Bayes, Random Forest), clustering (K-Means, hierarchical), neural networks and CNN, model evaluation and cross-validation. A paper that uses only advanced methods that were never taught (for example a very large transformer with its own

special training) is not allowed.

6. Application-Based Idea Examples

Choose an application that interests you. These examples only show the type of idea expected; you must find your own paper.

Application Area	**Example Problem**	**Typical Dataset Type**	**Example Methods**
Healthcare	Predicting diabetes or heart disease from patient records	Tabular medical data	Logistic Regression, Random Forest, SVM
Agriculture	Detecting plant leaf disease from images	Image dataset	CNN, transfer learning
Finance	Detecting credit card fraud	Imbalanced transactions	Decision Tree, XGBoost, resampling
Education	Predicting student performance or dropout	Student records	KNN, Naive Bayes, Random Forest
E-commerce	Customer segmentation	Purchase history	K-Means, hierarchical clustering
Social media / NLP	Fake news or sentiment detection	Text data	TF-IDF with SVM, LSTM
Traffic / Smart city	Traffic sign recognition	Image data	CNN
Cybersecurity	Network intrusion detection	Network logs	Random Forest, Neural Network

7. Deliverables

#	Deliverable	Format
1	Paper approval form (title, link, dataset link, reason)	1 page
2	Source code	Notebook (.ipynb) or .py files, well commented and link of the GITHUB
3	Project report	
5	Copy of the original paper	

8. Report Structure

1. **Title page:** project title, names, roll numbers, course, date.
2. **Abstract:** 150 to 200 words summarizing everything.
3. **Introduction:** application area and why the problem matters.
4. **Selected Paper Summary:** citation, problem statement, objectives, contributions.
5. **Dataset Description:** source, size, features, classes, EDA.
6. **Methodology:** pre-processing, models, parameters, evaluation protocol (include a flow diagram).
7. **Implementation Details:** tools, libraries, hardware, random seeds, assumptions.
8. **Results:** tables, graphs, confusion matrices.
9. **Comparison with the Paper:** comparison table and discussion of differences.
10. **Limitations and Future Work.**
11. **Conclusion.**
12. **References:** at least the paper plus 4 more sources, in IEEE or APA style.

9. Academic Integrity

- Your work must be your own. Copying code, results or text from the paper, from other students or from the internet without citation is **plagiarism** and will be penalized.
- Any use of AI tools for help must be stated honestly in the report, and you must be able to explain every line of your code and every part of your report during the viva.
- Cite every source you use.

10. Final Checklist

- Paper is from 2025 or 2026 and approved.
- Problem statement written clearly.
- Same dataset as the paper is used.
- Same methodology is applied, using course topics.
- Results recorded with the same metrics.
- Comparison table and discussion included.
- The code runs successfully from start to finish and is uploaded to GitHub.
- Report and paper copy submitted.
- References and integrity statement included.